

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		71073	
Total Threats Mitigated	43	Total Packets Analyzed		2822850904	
High Priority	12	Malicious Packets		2811768326	
Low Priority	31	True Negatives Passed		60756	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		18	
False Positives	128	UDP Flood		7	
FP Rate	0.15%	Uncertain		12	
IF Precision	99.05%				
IF Recall	93.57%	Action		Count	
IF F1-Score	96.23%	Quarantined		10	
IF Accuracy	98.94%	Time Ban		10	
		Blackhole		0	
		Released		5	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.05%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.57%	Share of actual attacks successfully flagged
F1-Score	96.23%	Balanced measure combining Precision and Recall
Accuracy	98.94%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.15%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.43%	Actual attacks that went undetected
True Positive Rate (TPR)	93.57%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.85%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 9513	FN 654
Actual: Normal	FP 91	TN 60102

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.98%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	2.20%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	97.80%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4341	0	0
Act: ICMP	0	1519	0
Act: UDP	0	0	3653

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.05%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.57%	Share of actual attacks successfully flagged
F1-Score	96.23%	Balanced measure combining Precision and Recall
Accuracy	98.94%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.15%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.43%	Actual attacks that went undetected
True Positive Rate (TPR)	93.57%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.85%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 9513	FN 654
Actual: Normal	FP 91	TN 60102

2d. Response Latency

Metric	Value	Description
Detection Time	422.0 ms	Average time taken to detect an attack
Mitigation Response Time	12.9 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.78%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	8.95%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	6.53%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	3	3	2026-09-25 12:08:47	2026-09-25 12:11:58
10.0.0.18	3	3	2026-09-25 12:08:47	2026-09-25 12:11:58
10.0.0.19	3	3	2026-09-25 12:08:47	2026-09-25 12:11:58
10.0.0.16	2	2	2026-09-25 12:12:04	2026-09-25 12:13:15
10.0.0.20	1	1	2026-09-25 12:12:34	2026-09-25 12:12:45
10.0.0.21	1	1	2026-09-25 12:12:34	2026-09-25 12:12:45
10.0.0.22	1	1	2026-09-25 12:12:35	2026-09-25 12:12:46
10.0.0.23	1	1	2026-09-25 12:13:04	2026-09-25 12:13:15

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	1	1	2026-09-25 12:13:05	2026-09-25 12:13:16
10.0.0.25	1	1	2026-09-25 12:13:04	2026-09-25 12:13:15
10.0.0.4	3	1	2026-09-25 12:12:52	2026-09-25 12:13:28
10.0.0.10	1	1	2026-09-25 12:12:57	2026-09-25 12:13:08

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:06:02	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:02	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:03	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 12:06:12	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:12	10.0.0.18	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-25 12:06:13	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-25 12:06:13	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-25 12:06:33	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:33	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:34	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	High	Time Ban (1m)
2026-09-25 12:06:43	10.0.0.20	DDoS	UDP Flood	75.6%	Critical	Time Ban
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-25 12:06:44	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-25 12:06:44	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban
2026-09-25 12:07:03	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:03	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:05	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 12:07:12	10.0.0.18	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:12	10.0.0.16	DDoS	SYN Flood	92.7%	High	Released
2026-09-25 12:07:13	10.0.0.17	DDoS	SYN Flood	93.3%	High	Time Ban (2m)
2026-09-25 12:07:13	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-25 12:07:13	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-25 12:07:14	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-25 12:07:14	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-25 12:07:15	10.0.0.25	DDoS	ICMP Flood	91.2%	Critical	Time Ban
2026-09-25 12:07:15	10.0.0.24	DDoS	ICMP Flood	91.2%	Critical	Time Ban (1m)
2026-09-25 12:07:16	10.0.0.24	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-25 12:07:23	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-25 12:09:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-25 12:11:58	10.0.0.19	DDoS	SYN Flood	92.7%	Critical	Time Ban (5m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 12:11:58	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:11:58	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-25 12:13:02	10.0.0.4	DDoS	Uncertain	59.6%	Low	Released
2026-09-25 12:13:02	10.0.0.9	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-25 12:13:08	10.0.0.10	DDoS	Uncertain	68.4%	Low	Released
2026-09-25 12:13:15	10.0.0.16	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.4	Uncertain	0.6125	60.4%	Low	Phase 1	1	10s	2026-09-25 12:13:28	Attack Stopped
10.0.0.24	ICMP Flood	0.6444	91.1%	High	Phase 2	1	11s	2026-09-25 12:13:16	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6286	92.5%	Critical	Phase 2	2	1m 11s	2026-09-25 12:13:15	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6454	90.8%	High	Phase 2	1	11s	2026-09-25 12:13:15	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	High	Phase 2	1	11s	2026-09-25 12:13:15	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6117	59.6%	Low	Phase 1	1	11s	2026-09-25 12:13:13	Attack Stopped
10.0.0.10	Uncertain	0.5880	68.4%	Low	Phase 1	1	11s	2026-09-25 12:13:08	Attack Stopped
10.0.0.4	Uncertain	0.6086	59.6%	Low	Phase 1	1	10s	2026-09-25 12:13:02	Attack Stopped
10.0.0.22	UDP Flood	0.6305	79.7%	High	Phase 2	1	11s	2026-09-25 12:12:46	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	79.2%	High	Phase 2	1	11s	2026-09-25 12:12:45	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	83.9%	High	Phase 2	1	11s	2026-09-25 12:12:45	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6371	92.5%	Critical	Phase 2	1	10s	2026-09-25 12:12:14	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6305	92.7%	Critical	Phase 2	3	3m 11s	2026-09-25 12:11:58	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6258	93.4%	Critical	Phase 2	3	3m 11s	2026-09-25 12:11:58	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6413	93.4%	Critical	Phase 2	3	3m 11s	2026-09-25 12:11:58	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6305	92.7%	Critical	Phase 2	2	1m 11s	2026-09-25 12:09:58	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6258	93.4%	Critical	Phase 2	2	1m 11s	2026-09-25 12:09:58	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6413	93.4%	Critical	Phase 2	2	1m 11s	2026-09-25 12:09:58	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6305	92.7%	Critical	Phase 2	1	11s	2026-09-25 12:08:58	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6258	93.4%	Critical	Phase 2	1	11s	2026-09-25 12:08:58	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6413	93.4%	Critical	Phase 2	1	11s	2026-09-25 12:08:58	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE